

MuSc: Zero-Shot Industrial Anomaly Classification & Segmentation with Mutual Scoring of the Unlabeled Images

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Problem

Industrial anomaly classification (AC) and segmentation (AS) usually need many normal reference images (one-class methods) or hand-crafted text prompts (CLIP-based zero-shot methods) — both limit real-world deployment.

A true zero-shot setting has no training data, no text prompts, and no normal reference images, yet defects must still be flagged per-image and localized per-pixel.

Motivation

- Prior methods ignore the abundant normal and abnormal cues already sitting inside the unlabeled test images themselves.
- Those cues alone are enough to determine anomalies with no labels, prompts, or normal references.

A normal patch finds many similar patches across the other unlabeled images; an abnormal patch finds only a few — a discriminative, training-free signal.

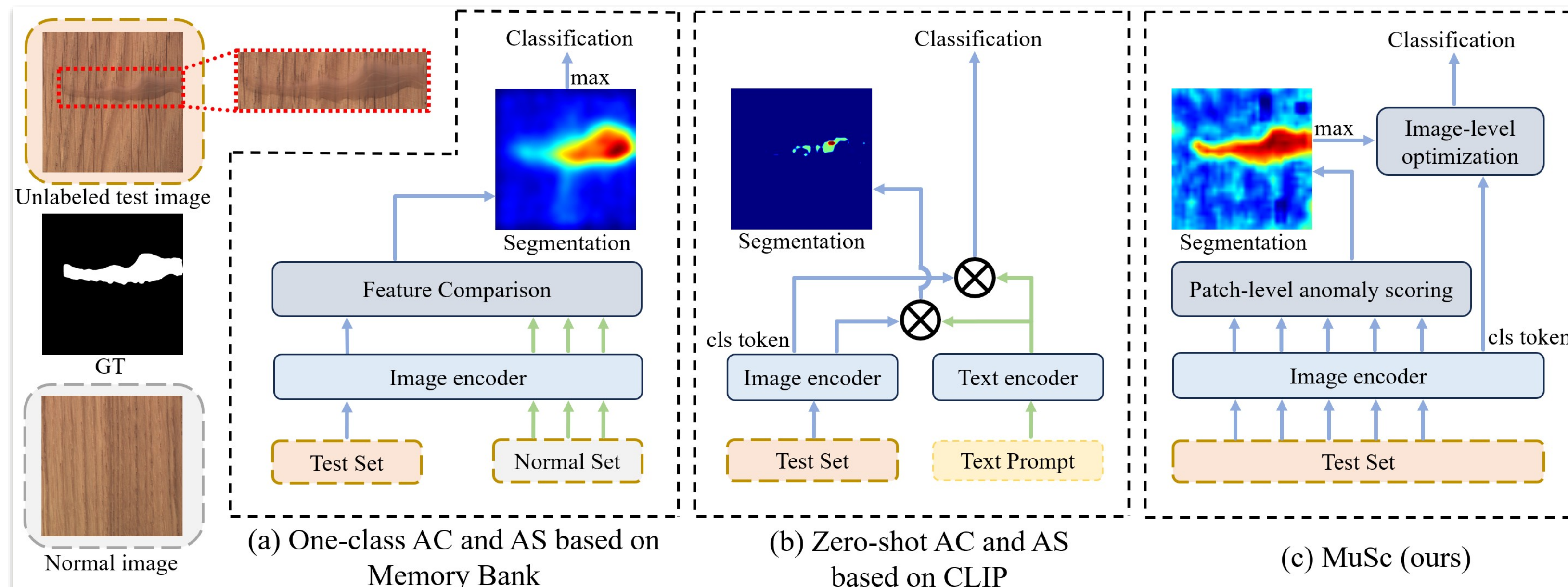


Fig. 1 · Prior paradigms need normal reference images (a) or text prompts (b); MuSc (c) leverages only the unlabeled test images.

Method

- LNAMD** aggregates frozen ViT patch tokens at multiple degrees r for multi-size anomalies.
- MSM** mutually scores patches across test images; **RsCIN** refines the image-level decision.

$$s_i = \frac{1}{K} \sum_{k=1}^K \min_n \|\hat{p}_i - \hat{p}_{k,n}\|_2$$

MSM score: mean nearest-patch distance over the K most-similar test images.

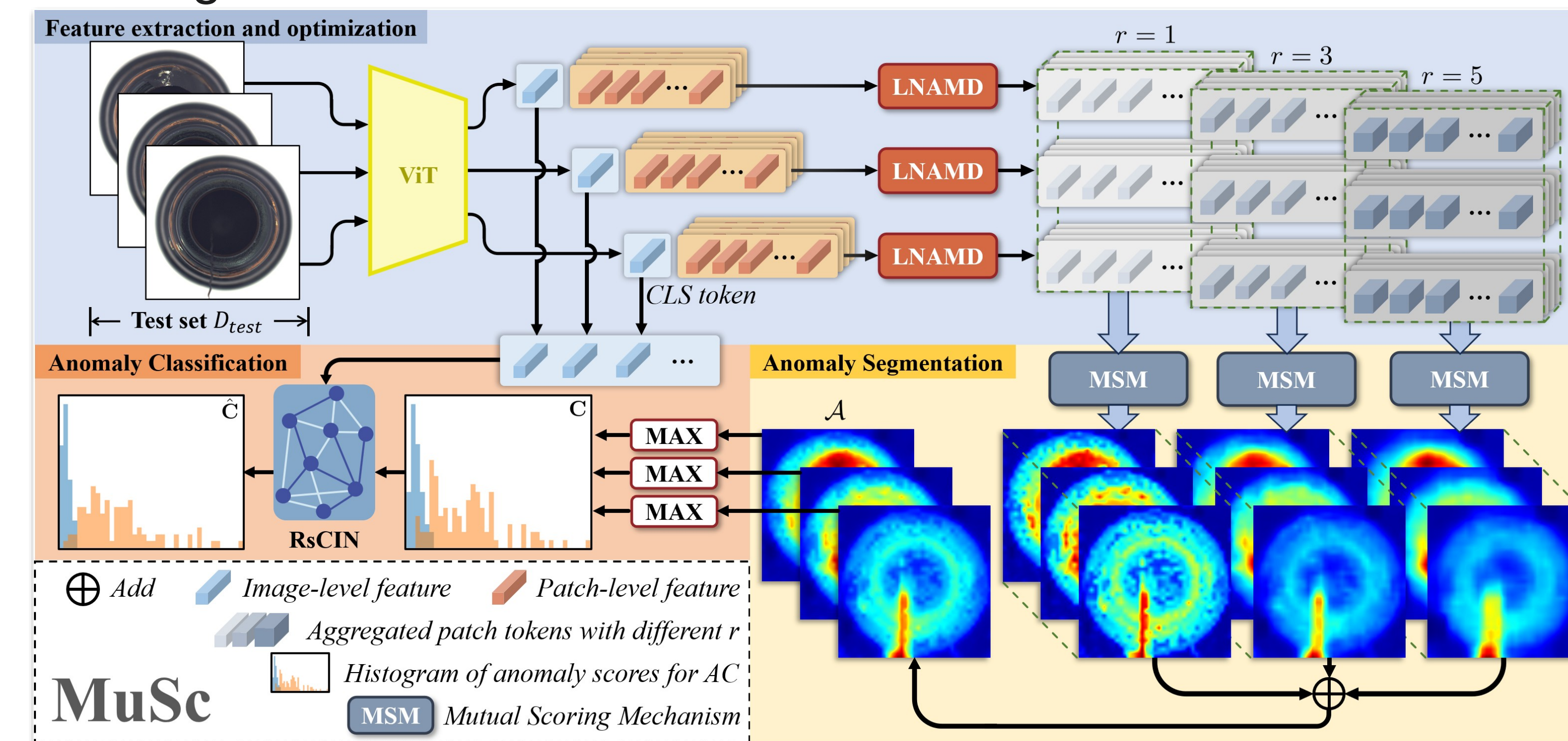


Fig. 2 · MuSc architecture: LNAMD → MSM (segmentation) → RsCIN (classification).

Ablation Study

- All three degrees $r \in \{1, 3, 5\}$ in LNAMD give the best combined AC/AS: small r for tiny defects, large r for big ones.
- Averaging the minimum 30% interval in MSM reaches **97.8%** AC vs 83.8% for a max strategy; RsCIN lifts VisAAC from 90.0% to 92.8%.

Key Results

0-shot method	AC AUROC	AS AUROC
WinCLIP	91.8	85.1
MuSc (ours)	97.8	97.3

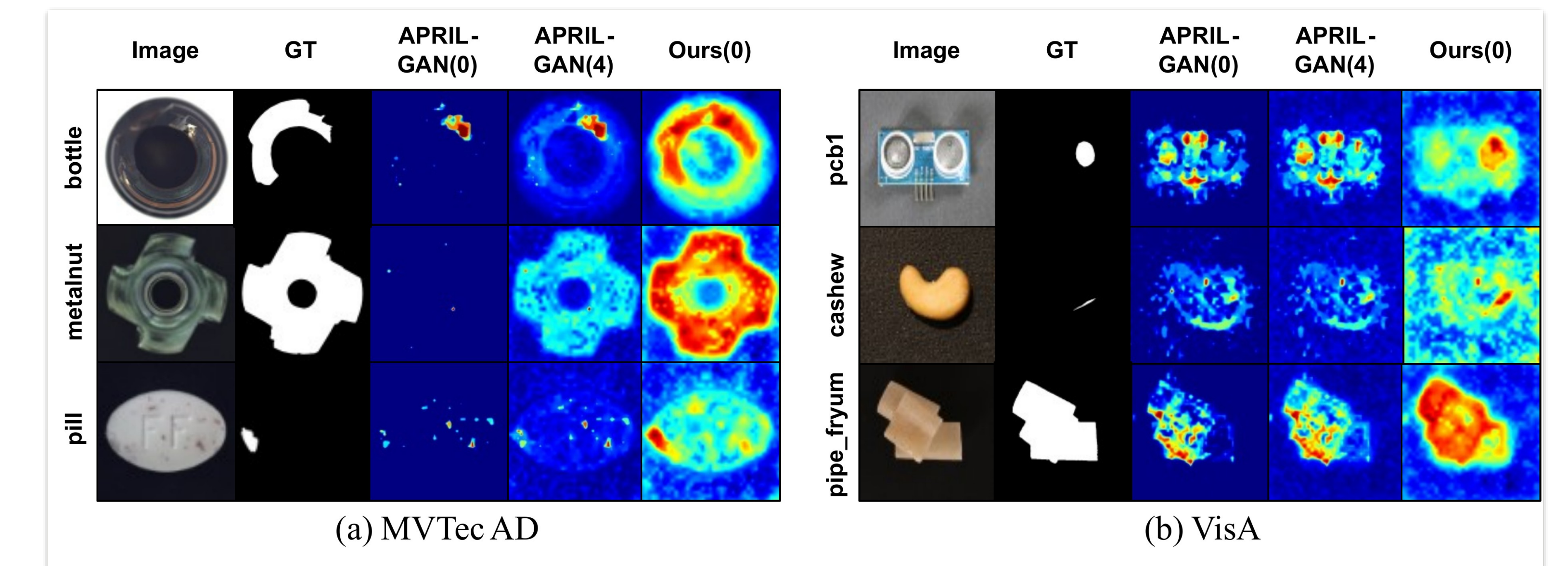


Fig. 6 · Zero-shot segmentation on MVTec AD & VisA.

Headline Numbers

97.8%	97.3% pixel-AUROC (MVTec AD)
image-AUROC, MVTec AD (0-shot)	92.8% image-AUROC (VisA)
<i>new zero-shot SOTA — no training or prompts</i>	+21.1 PRO gain vs 2nd-best

Takeaway

The lasting idea: a batch of unlabeled test images can supervise itself. Letting the images score one another yields state-of-the-art zero-shot classification and segmentation with no training, prompts, or reference images — even rivalling few-shot and full-shot methods that use the entire normal set, all with no data collection or annotation.